Class 12 Mathematics

Chapter 6: Application of Derivatives

Practice Set B: step-wise solutions

20 questions **52** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** Board-pattern — 1 mark

The radius of a right circular cylinder is increasing at the rate of 3 cm/s and its height is decreasing at the rate of 4 cm/s. The rate of change of its volume when the radius is 4 cm and the height is 6 cm is:

(A) $80\pi \text{ cm}^3/\text{s}$ **(B)** $144\pi \text{ cm}^3/\text{s}$ **(C)** $208\pi \text{ cm}^3/\text{s}$ **(D)** $64\pi \text{ cm}^3/\text{s}$

Formula / idea: Both r and h change with t, so use the product rule on $V = \pi r^2 h$.

1 Differentiate $V = \pi r^2 h$ w.r.t. t (product rule)

$$\frac{dV}{dt} = \pi\left(2rh\,\frac{dr}{dt} + r^2\,\frac{dh}{dt}\right)$$

2 Put $r = 4$, $h = 6$, $\frac{dr}{dt} = 3$, $\frac{dh}{dt} = -4$ ✓ 1

$$\frac{dV}{dt} = \pi\big(2(4)(6)(3) + 16(-4)\big) = \pi(144 - 64) = 80\pi$$

Height is decreasing, so $\frac{dh}{dt}$ is negative

Answer **(A)** $80\pi \text{ cm}^3/\text{s}$

Watch out: Taking $\frac{dh}{dt} = +4$ gives 208π; dropping the h-term gives 144π.

2 **MCQ** NCERT Exemplar — 1 mark

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x + \cos x$. Then f:

(A) has a minimum at $x = \pi$ **(B)** has a maximum at $x = 0$

(C) is a decreasing function **(D)** **is an increasing function**

Formula / idea: $f'(x) > 0$ for all x means f is increasing on $\mathbb{R}$ and has no local extremum.

1 Differentiate

$$f'(x) = 2 - \sin x$$

2 Since $\sin x \leq 1$, $f'(x) > 0$ for every real x ✓ 1

$f'(x) = 2 - \sin x \geq 2 - 1 = 1 > 0$

3 So f is increasing on $\mathbb{R}$, with no maximum or minimum

Option (D)

Answer **(D) is an increasing function**

Watch out: Thinking $\cos x$ has a minimum at π, so f must too; the $2x$ term keeps f rising.

3 **MCQ** CBSE SQP 2026-27 1 mark

If $y = a\left\{x + 2a\sin\left(\frac{x}{a}\right)\right\}$, $a > 0$, $x \in [0, \pi a]$, is a decreasing function, then the interval in which x lies is:

(A) $\left(\frac{5\pi}{6}, \pi\right]$ **(B)** $\left(\frac{2\pi}{3}, \pi\right]$ **(C)** $\left(\frac{5\pi a}{6}, \pi a\right]$ **(D)** $\left(\frac{2\pi a}{3}, \pi a\right]$

Formula / idea: y is decreasing where $\frac{dy}{dx} < 0$.

1 Differentiate w.r.t. x

$\frac{dy}{dx} = a\left\{1 + 2\cos\left(\frac{x}{a}\right)\right\}$

2 Decreasing needs $\frac{dy}{dx} < 0$; since $a > 0$

$\cos\left(\frac{x}{a}\right) < -\frac{1}{2}$

3 Here $\frac{x}{a} \in [0, \pi]$, where $\cos\theta < -\frac{1}{2}$ for $\theta \in \left(\frac{2\pi}{3}, \pi\right]$ ✓ 1

$\frac{2\pi}{3} < \frac{x}{a} \leq \pi \Rightarrow \frac{2\pi a}{3} < x \leq \pi a$

Answer **(D)** $\left(\frac{2\pi a}{3}, \pi a\right]$

Watch out: Stopping at $\frac{x}{a}$ and choosing $\left(\frac{2\pi}{3}, \pi\right]$; multiply back by a.

4 MCQ NCERT Exemplar 1 mark

At $x = \frac{5\pi}{6}$, $f(x) = 2\sin 3x + 3\cos 3x$ is:

(A) maximum **(B)** minimum

(C) zero **(D)** **neither maximum nor minimum**

Formula / idea: A local maximum or minimum of a differentiable function needs $f'(c) = 0$.

1 Value at $x = \frac{5\pi}{6}$ $(3x = \frac{5\pi}{2})$

$$f\left(\frac{5\pi}{6}\right) = 2\sin\frac{5\pi}{2} + 3\cos\frac{5\pi}{2} = 2(1) + 0 = 2 \neq 0$$

2 Differentiate

$$f'(x) = 6\cos 3x - 9\sin 3x$$

3 $f'\left(\frac{5\pi}{6}\right) \neq 0$, so it is not a point of extremum ✓ 1

$$f'\left(\frac{5\pi}{6}\right) = 6(0) - 9(1) = -9 \neq 0$$

Answer **(D) neither maximum nor minimum**

Watch out: Taking $\sin 3x = 1$ as a sign of a maximum; the maximum of f is $\sqrt{13}$, not 2.

5 MCQ Board-pattern 1 mark

The local minimum value of $f(x) = x^x$, $x > 0$, is:

(A) $e^{-1/e}$ **(B)** $e^{1/e}$ **(C)** $\frac{1}{e}$ **(D)** 1

Formula / idea: For $y = x^x$ take logs first: $\log y = x\log x$.

1 Take logs and differentiate w.r.t. x

$$\log y = x\log x \;\Rightarrow\; \frac{1}{y}\frac{dy}{dx} = 1 + \log x$$

2 So $f'(x) = x^x(1 + \log x) = 0$ gives

$$\log x = -1 \;\Rightarrow\; x = \frac{1}{e}$$

3 f' changes from $-$ to $+$ at $x = \frac{1}{e}$, so it is a local minimum

$$x^x > 0, \quad 1 + \log x < 0 \text{ for } x < \tfrac{1}{e},\ > 0 \text{ for } x > \tfrac{1}{e}$$

4 Local minimum value

$$f\left(\frac{1}{e}\right) = \left(\frac{1}{e}\right)^{1/e} = e^{-1/e}$$

✓ 1

Answer **(A)** $e^{-1/e}$

Watch out: Giving the point $x = \frac{1}{e}$ instead of the value $f\left(\frac{1}{e}\right)$.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): $f(x) = x^3$ is strictly increasing on $\mathbb{R}$. **Reason (R):** If a function f is strictly increasing on an interval, then $f'(x) > 0$ for every x in that interval.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) **Assertion (A) is true, but Reason (R) is false.**

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: $f' > 0$ on an interval implies strictly increasing, but the converse is false.

1 $f'(x) = 3x^2 > 0$ except at $x = 0$, so f is strictly increasing: **A is true**

$x_1 < x_2 \Rightarrow x_1^3 < x_2^3$

2 x^3 is strictly increasing, yet $f'(0) = 0$: **R is false**

$f'(0) = 3(0)^2 = 0$

3 A is true and R is false

Option (C)

✓ 1

Answer **(C) A is true, but R is false.**

Watch out: Choosing (A): R is the converse of the theorem, and x^3 is the counter-example.

7 **Short answer** NCERT Ex 6.1 Q2 2 marks

The volume of a cube is increasing at the rate of $8\ \text{cm}^3/\text{s}$. How fast is the surface area increasing when the length of an edge is 12 cm?

Formula / idea: Find $\frac{dx}{dt}$ from V, then use it in $\frac{dS}{dt}$.

1 Let edge $= x$; volume and surface area

$V = x^3, \qquad S = 6x^2$

✓ ½

2 Differentiate V w.r.t. t and use $\dfrac{dV}{dt} = 8$

$$3x^2 \frac{dx}{dt} = 8 \;\Rightarrow\; \frac{dx}{dt} = \frac{8}{3x^2}$$

✓ ½

3 Differentiate S w.r.t. t

$$\frac{dS}{dt} = 12x \frac{dx}{dt} = 12x \cdot \frac{8}{3x^2} = \frac{32}{x}$$

✓ ½

4 Put $x = 12$

$$\frac{dS}{dt} = \frac{32}{12} = \frac{8}{3}\ \text{cm}^2/\text{s}$$

✓ ½

Answer **Surface area increases at $\dfrac{8}{3}\ \text{cm}^2/\text{s}$**

Watch out: Using $\dfrac{dx}{dt} = 8$; the 8 is the rate of change of volume, not of the edge.

Level 2

19 marks

8 **Short answer** PYQ-style 3 marks

Find the intervals in which the function $f(x) = \dfrac{x^4}{4} - x^3 - 5x^2 + 24x + 12$ is (a) strictly increasing (b) strictly decreasing.

Formula / idea: Sign of $f'(x)$ on each interval between the roots of $f'(x) = 0$.

1 Differentiate

$$f'(x) = x^3 - 3x^2 - 10x + 24$$

✓ ½

2 $x = 2$ is a root ($8 - 12 - 20 + 24 = 0$); factorise

$$f'(x) = (x-2)(x^2 - x - 12) = (x-2)(x-4)(x+3)$$

✓ ½

3 $f'(x) = 0$ gives the points

$$x = -3,\ 2,\ 4$$

✓ ½

4 Signs of $f'(x)$ on the intervals

$$(-\infty, -3) : (-)(-)(-) < 0$$
$$(-3, 2) : (-)(-)(+) > 0$$
$$(2, 4) : (+)(-)(+) < 0$$
$$(4, \infty) : (+)(+)(+) > 0$$

✓ 1

5 (a) Strictly increasing where $f'(x) > 0$; (b) strictly decreasing where $f'(x) < 0$

(a) $(-3, 2) \cup (4, \infty)$ (b) $(-\infty, -3) \cup (2, 4)$

✓ ½

Answer **Strictly increasing on $(-3, 2) \cup (4, \infty)$; strictly decreasing on $(-\infty, -3) \cup (2, 4)$**

Watch out: Not finding the root $x = 2$ by trial; test $x = \pm 1, \pm 2, \pm 3, \ldots$ in $f'(x)$.

9 **Short answer** PYQ-style 2 marks

A spherical balloon is being inflated so that its volume increases at the rate of $25\ \text{cm}^3/\text{s}$. Find the rate at which its surface area is increasing when the radius is 5 cm.

Formula / idea: $V = \frac{4}{3}\pi r^3$ and $S = 4\pi r^2$; link them through $\frac{dr}{dt}$.

1	Differentiate $V = \frac{4}{3}\pi r^3$ w.r.t. t $\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$	✓ ½
2	Put $\frac{dV}{dt} = 25$, $r = 5$ $25 = 4\pi(25)\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{1}{4\pi}$	✓ ½
3	Differentiate $S = 4\pi r^2$ w.r.t. t $\frac{dS}{dt} = 8\pi r \frac{dr}{dt}$	✓ ½
4	Put $r = 5$, $\frac{dr}{dt} = \frac{1}{4\pi}$ $\frac{dS}{dt} = 8\pi(5)\cdot\frac{1}{4\pi} = 10\ \text{cm}^2/\text{s}$	✓ ½

Answer **Surface area increases at** $10\ \text{cm}^2/\text{s}$

Watch out: Writing $\frac{dS}{dt} = 8\pi r$ and forgetting the factor $\frac{dr}{dt}$.

10 **Short answer** NCERT Exemplar 3 marks

Find the points of local maxima, local minima and the points of inflexion (critical points where f' does not change sign) of the function $f(x) = x^5 - 5x^4 + 5x^3 - 1$. Also find the corresponding local maximum and local minimum values.

Formula / idea: First derivative test: f' from $+$ to $-$ is a maximum; no sign change is a point of inflexion.

1	Differentiate and factorise $f'(x) = 5x^4 - 20x^3 + 15x^2 = 5x^2(x-1)(x-3)$	✓ ½
2	$f'(x) = 0$ gives $x = 0,\ 1,\ 3$	✓ ½
3	Near $x = 0$, $f' > 0$ on both sides: no sign change $x = 0$ is a point of inflexion	✓ ½
4	At $x = 1$: f' changes from $+$ to $-$; at $x = 3$: from $-$ to $+$ $x = 1:$ local maximum, $\quad x = 3:$ local minimum	✓ 1

5 Corresponding values
$f(1) = 1 - 5 + 5 - 1 = 0, \qquad f(3) = 243 - 405 + 135 - 1 = -28$ ✓ ½

Answer **Local maximum at $x = 1$, value 0; local minimum at $x = 3$, value -28; point of inflexion at $x = 0$**

Watch out: Calling $x = 0$ a maximum or minimum just because $f'(0) = 0$; check the sign change.

11 **Short answer** PYQ-style 3 marks

Find the intervals in which the function $f(x) = \log(1 + x) - \dfrac{x}{1 + x}$, $x > -1$, is strictly increasing or strictly decreasing.

Formula / idea: On the domain $x > -1$, the sign of $f'(x)$ decides increasing or decreasing.

1 Differentiate (quotient rule for the second term)
$f'(x) = \dfrac{1}{1 + x} - \dfrac{(1 + x) - x}{(1 + x)^2}$ ✓ 1

2 Simplify
$f'(x) = \dfrac{1}{1 + x} - \dfrac{1}{(1 + x)^2} = \dfrac{x}{(1 + x)^2}$ ✓ ½

3 $(1 + x)^2 > 0$, so $f'(x)$ has the sign of x; $f'(x) = 0$ at $x = 0$
$-1 < x < 0 : f'(x) < 0, \qquad x > 0 : f'(x) > 0$ ✓ 1

4 Hence
strictly decreasing on $(-1, 0)$, strictly increasing on $(0, \infty)$ ✓ ½

Answer **Strictly decreasing on $(-1, 0)$; strictly increasing on $(0, \infty)$**

Watch out: Writing $(-\infty, 0)$ for the decreasing interval; the domain starts at $x = -1$.

12 **Short answer** CBSE SQP 2026-27 3 marks

The circular area of vision on the ground by a surveillance drone fitted with a camera increases at the rate of $44\ \text{m}^2$ per metre as it gains vertical height. The radius of the circle on the ground increases at the rate of 6 m/s. Find the rate at which the height of the drone increases with time, at the instant when the radius of the circular area on the ground is 14 m. (Use $\pi = \dfrac{22}{7}$)

Formula / idea: Chain rule: $\dfrac{dA}{dt} = \dfrac{dA}{dh} \cdot \dfrac{dh}{dt}$.

1 Let area $= A$, radius $= r$, height $= h$ at time t; given
$\dfrac{dA}{dh} = 44\ \text{m}^2/\text{m}, \qquad \dfrac{dr}{dt} = 6\ \text{m/s}$ ✓ ½

2	Area $A = \pi r^2$; differentiate w.r.t. t $\dfrac{dA}{dt} = 2\pi r \dfrac{dr}{dt}$	✓ ½
3	Put $r = 14$, $\dfrac{dr}{dt} = 6$ $\dfrac{dA}{dt} = 2 \times \dfrac{22}{7} \times 14 \times 6 = 528\ \text{m}^2/\text{s}$	✓ ½
4	By the chain rule $\dfrac{dA}{dt} = \dfrac{dA}{dh} \cdot \dfrac{dh}{dt} \Rightarrow 528 = 44 \dfrac{dh}{dt}$	✓ 1
5	Hence $\dfrac{dh}{dt} = \dfrac{528}{44} = 12\ \text{m/s}$	✓ ½

Answer **The height of the drone increases at 12 m/s**

Watch out: Treating 44 as $\dfrac{dA}{dt}$; it is the rate per metre of height, $\dfrac{dA}{dh}$.

13 Short answer NCERT Misc. Ex. Q11 — 3 marks

Find the absolute maximum and minimum values of the function f given by $f(x) = \cos^2 x + \sin x$, $x \in [0, \pi]$.

Formula / idea: Compare f at the critical points and at the end points of $[0, \pi]$.

1	Differentiate $f'(x) = -2\cos x \sin x + \cos x = \cos x\,(1 - 2\sin x)$	✓ ½
2	$f'(x) = 0$ in $[0, \pi]$ $\cos x = 0 \Rightarrow x = \dfrac{\pi}{2}$; $\quad \sin x = \dfrac{1}{2} \Rightarrow x = \dfrac{\pi}{6}, \dfrac{5\pi}{6}$	✓ 1
3	Values at critical points and end points $f(0) = 1, \quad f\left(\frac{\pi}{6}\right) = \frac{3}{4} + \frac{1}{2} = \frac{5}{4}, \quad f\left(\frac{\pi}{2}\right) = 0 + 1 = 1$ $f\left(\frac{5\pi}{6}\right) = \frac{3}{4} + \frac{1}{2} = \frac{5}{4}, \quad f(\pi) = 1 + 0 = 1$	✓ 1
4	Compare the values $\text{Absolute max} = \dfrac{5}{4}$ at $x = \dfrac{\pi}{6}, \dfrac{5\pi}{6}$; $\quad \text{Absolute min} = 1$ at $x = 0, \dfrac{\pi}{2}, \pi$	✓ ½

Answer **Absolute maximum $= \dfrac{5}{4}$ (at $x = \dfrac{\pi}{6}, \dfrac{5\pi}{6}$); absolute minimum $= 1$ (at $x = 0, \dfrac{\pi}{2}, \pi$)**

Watch out: Missing $x = \dfrac{5\pi}{6}$: $\sin x = \dfrac{1}{2}$ has two solutions in $[0, \pi]$.

14 **Short answer** NCERT Ex 6.3 Q14 2 marks

Find two positive numbers x and y such that $x + y = 60$ and xy^3 is maximum.

Formula / idea: Write the product in one variable, then $P' = 0$ and $P'' < 0$.

1	Put $x = 60 - y$; let $P = xy^3$ $P = (60 - y)y^3 = 60y^3 - y^4$	✓ ½
2	Differentiate and put $\frac{dP}{dy} = 0$ $\frac{dP}{dy} = 180y^2 - 4y^3 = 4y^2(45 - y) = 0 \Rightarrow y = 45$ $y > 0$, so $y \neq 0$	✓ ½
3	Second derivative test $\frac{d^2P}{dy^2} = 360y - 12y^2 = 16200 - 24300 = -8100 < 0$	✓ ½
4	So P is maximum at $y = 45$; hence the numbers are $x = 60 - 45 = 15, \quad y = 45$	✓ ½

Answer $x = 15, y = 45$

Watch out: Writing P in terms of x and then mixing up which number is 15 and which is 45.

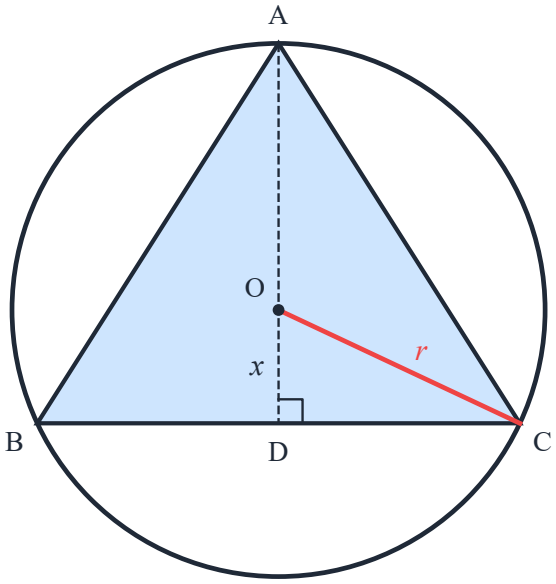
Level 3

25 marks

15 **Long answer** PYQ-style 5 marks

Show that the isosceles triangle of maximum area that can be inscribed in a circle of radius r is an equilateral triangle. Also find its maximum area.

Formula / idea: Area and (area)^2 are maximum together, so maximise $S = A^2$.



1	Let $AB = AC$; altitude AD passes through centre O; let $OD = x$ $AD = r + x, \qquad BD = \sqrt{OB^2 - OD^2} = \sqrt{r^2 - x^2}$	✓ 1
2	Area of the triangle $A = \frac{1}{2} \cdot BC \cdot AD = (r + x)\sqrt{r^2 - x^2}$	✓ ½
3	Let $S = A^2$ $S = (r + x)^2(r^2 - x^2) = (r + x)^3(r - x)$ $A > 0$, so A and A^2 are maximum together	✓ ½
4	Differentiate w.r.t. x $\frac{dS}{dx} = 3(r + x)^2(r - x) - (r + x)^3 = (r + x)^2(2r - 4x)$	✓ ½
5	Put $\frac{dS}{dx} = 0$ $(r + x \neq 0)$ $x = \frac{r}{2}$	✓ ½
6	Second derivative at $x = \frac{r}{2}$ $\frac{d^2S}{dx^2} = 2(r + x)(2r - 4x) - 4(r + x)^2 = 0 - 4 \cdot \frac{9r^2}{4} = -9r^2 < 0$	✓ 1

7 So the area is maximum at $x = \dfrac{r}{2}$; find the sides ✓ ½

$AD = \frac{3r}{2}, \quad BD = \frac{\sqrt{3}\,r}{2} \;\Rightarrow\; BC = \sqrt{3}\,r$

$AB = \sqrt{AD^2 + BD^2} = \sqrt{\frac{9r^2}{4} + \frac{3r^2}{4}} = \sqrt{3}\,r$

8 $AB = AC = BC$: the triangle is equilateral; maximum area ✓ ½

$$A = AD \cdot BD = \frac{3r}{2} \cdot \frac{\sqrt{3}\,r}{2} = \frac{3\sqrt{3}}{4} r^2$$

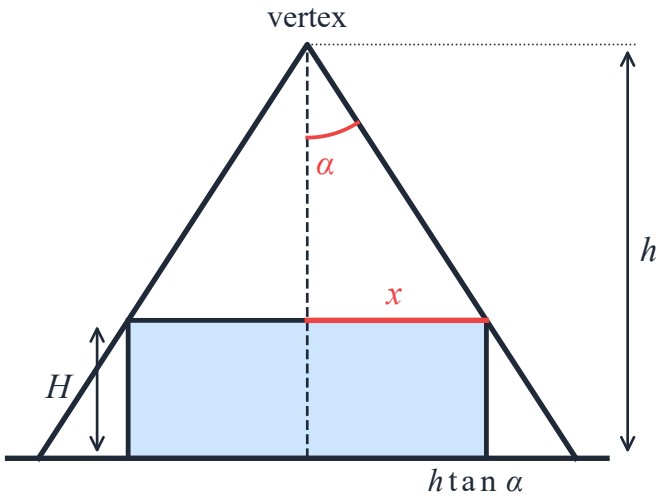
Answer **Maximum area when all sides are $\sqrt{3}\,r$ (equilateral); maximum area $= \dfrac{3\sqrt{3}}{4} r^2$**

Watch out: Writing $AD = r - x$; the vertex A is on the far side of O from BC, so $AD = r + x$.

16 **Long answer** NCERT Misc. Ex. Q15 5 marks

Show that the height of the cylinder of greatest volume which can be inscribed in a right circular cone of height h and semi-vertical angle α is one-third that of the cone and the greatest volume of the cylinder is $\dfrac{4}{27}\pi h^3 \tan^2 \alpha$.

Formula / idea: Use similar triangles to write the cylinder's height in terms of its radius, then maximise V.



1 Let the cylinder have radius x and height H; the cone's radius at height H is $(h - H)\tan\alpha$ ✓ 1

$x = (h - H)\tan\alpha \;\Rightarrow\; H = h - x\cot\alpha$

2 Volume of the cylinder ✓ ½

$V = \pi x^2 H = \pi\left(hx^2 - x^3 \cot\alpha\right)$

3 Differentiate w.r.t. x ✓ ½

$$\frac{dV}{dx} = \pi\left(2hx - 3x^2 \cot\alpha\right)$$

4 Put $\dfrac{dV}{dx} = 0\ (x \neq 0)$

$$x = \frac{2h}{3\cot\alpha} = \frac{2h}{3}\tan\alpha$$

✓ ½

5 Second derivative test

$$\frac{d^2V}{dx^2} = \pi\,(2h - 6x\cot\alpha) = \pi(2h - 4h) = -2\pi h < 0$$

✓ 1

6 So V is greatest; height of the cylinder

$$H = h - \frac{2h}{3}\tan\alpha\cot\alpha = h - \frac{2h}{3} = \frac{h}{3}$$

✓ ½

7 Greatest volume

$$V = \pi \cdot \frac{4h^2\tan^2\alpha}{9} \cdot \frac{h}{3} = \frac{4}{27}\pi h^3\tan^2\alpha$$

✓ 1

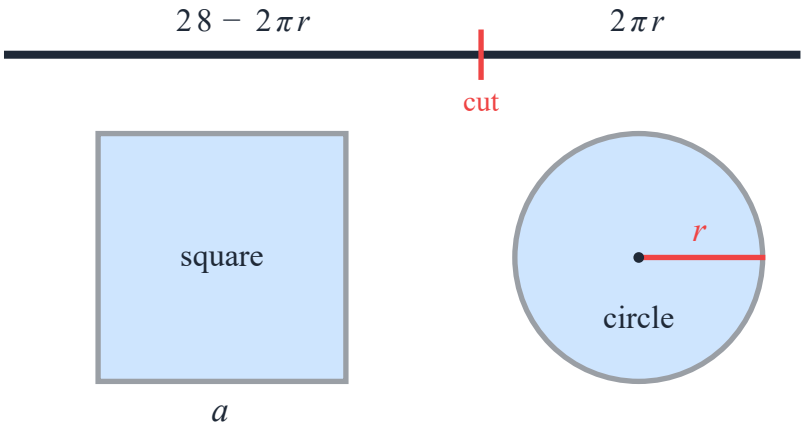
Answer **Height of cylinder** $= \dfrac{h}{3}$**; greatest volume** $= \dfrac{4}{27}\pi h^3\tan^2\alpha$

Watch out: Writing $H = x\cot\alpha$; that is the cone's height above the cylinder, not H.

17 **Long answer** NCERT Ex 6.3 Q22 5 marks

A wire of length 28 m is to be cut into two pieces. One of the pieces is to be made into a square and the other into a circle. What should be the length of the two pieces so that the combined area of the square and the circle is minimum?

Formula / idea: Write the total area in one variable using $4a + 2\pi r = 28$; minimum needs $A' = 0$, $A'' > 0$.



1 Let the square have side a and the circle radius r; total length

$$4a + 2\pi r = 28 \;\Rightarrow\; a = 7 - \frac{\pi r}{2}$$

✓ 1

2 Combined area

$$A = a^2 + \pi r^2 = \left(7 - \frac{\pi r}{2}\right)^2 + \pi r^2$$

✓ ½

3 Differentiate w.r.t. r ✓ 1

$$\frac{dA}{dr} = 2\left(7 - \frac{\pi r}{2}\right)\left(-\frac{\pi}{2}\right) + 2\pi r = -7\pi + \frac{\pi^2 r}{2} + 2\pi r$$

4 Put $\frac{dA}{dr} = 0$ and divide by π ✓ ½

$$\frac{\pi r}{2} + 2r = 7 \;\Rightarrow\; r = \frac{14}{\pi + 4}$$

5 Second derivative test ✓ 1

$$\frac{d^2A}{dr^2} = \frac{\pi^2}{2} + 2\pi > 0$$

6 So A is minimum; lengths of the two pieces ✓ 1

$$\text{circle: } 2\pi r = \frac{28\pi}{\pi + 4}\ \text{m}$$

$$\text{square: } 28 - \frac{28\pi}{\pi + 4} = \frac{112}{\pi + 4}\ \text{m}$$

Answer **Square piece** $= \frac{112}{\pi + 4}$ **m and circle piece** $= \frac{28\pi}{\pi + 4}$ **m**

Watch out: Giving r or a as the answer; the question asks for the lengths of the two pieces of wire.

18 Short answer NCERT Misc. Ex. Q3 — 3 marks

Find the intervals in which the function f given by $f(x) = \frac{4\sin x - 2x - x\cos x}{2 + \cos x}$, $0 \le x \le 2\pi$, is (i) increasing (ii) decreasing.

Formula / idea: Split $f(x) = \frac{4\sin x}{2 + \cos x} - x$ before differentiating.

1 Split the fraction ✓ ½

$$f(x) = \frac{4\sin x - x(2 + \cos x)}{2 + \cos x} = \frac{4\sin x}{2 + \cos x} - x$$

2 Differentiate (quotient rule) ✓ ½

$$f'(x) = \frac{4\cos x(2 + \cos x) + 4\sin^2 x}{(2 + \cos x)^2} - 1 = \frac{8\cos x + 4}{(2 + \cos x)^2} - 1$$

3 Simplify ✓ ½

$$f'(x) = \frac{8\cos x + 4 - 4 - 4\cos x - \cos^2 x}{(2 + \cos x)^2} = \frac{\cos x\,(4 - \cos x)}{(2 + \cos x)^2}$$

4 $4 - \cos x > 0$ and $(2 + \cos x)^2 > 0$, so $f'(x)$ has the sign of $\cos x$ ✓ ½

$$f'(x) = 0 \Rightarrow \cos x = 0 \Rightarrow x = \frac{\pi}{2}, \frac{3\pi}{2}$$

5 Sign of $\cos x$ on the intervals ✓ ½

$\left(0, \frac{\pi}{2}\right) : +, \qquad \left(\frac{\pi}{2}, \frac{3\pi}{2}\right) : -, \qquad \left(\frac{3\pi}{2}, 2\pi\right) : +$

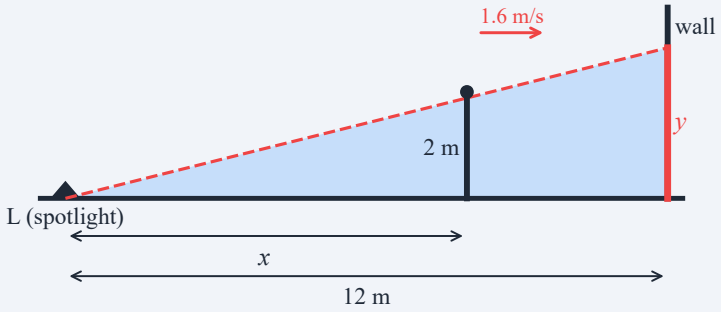
6 Hence
(i) increasing on $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right)$, (ii) decreasing on $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ ✓ ½

Answer **(i) Increasing on** $\left(0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 2\pi\right)$ **(ii) decreasing on** $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$

Watch out: Applying the quotient rule to the whole fraction and getting lost; split off $-x$ first.

19 **Short answer** Board-pattern 3 marks

A spotlight on the ground shines on a vertical wall 12 m away. A man 2 m tall walks from the spotlight straight towards the wall at a speed of 1.6 m/s. How fast is the height of his shadow on the wall decreasing when he is 4 m from the wall?



Formula / idea: Similar triangles give y in terms of x; then differentiate w.r.t. t.

1 Let the man be x m from the spotlight and the shadow be y m high; similar triangles
$\frac{y}{12} = \frac{2}{x} \Rightarrow y = \frac{24}{x}$ ✓ 1

2 Differentiate w.r.t. t
$\frac{dy}{dt} = -\frac{24}{x^2}\frac{dx}{dt}$ ✓ 1

3 4 m from the wall means $x = 12 - 4 = 8$; put $\frac{dx}{dt} = 1.6$
$\frac{dy}{dt} = -\frac{24}{64} \times 1.6 = -0.6$ ✓ ½

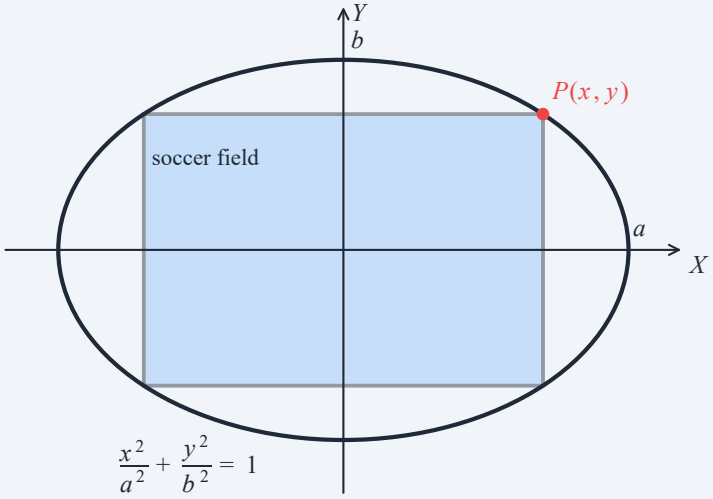
4 Negative sign: the shadow is getting shorter
Shadow decreases at 0.6 m/s ✓ ½

Answer **The shadow on the wall is decreasing at** 0.6 **m/s**

Watch out: Putting $x = 4$; x is measured from the spotlight, so $x = 12 - 4 = 8$.

20 **Case study** CBSE SQP 2022-23 4 marks

Case Study: In an elliptical sports field, the authority wants to design a rectangular soccer field with the maximum possible area. The sports field is given by the graph of $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.



(i) If the length and the breadth of the rectangular field are $2x$ and $2y$ respectively, find the area function in terms of x. 1 mark

(ii) Find the critical point of the function. 1 mark

(iii)(a) Use the first derivative test to find the length $2x$ and width $2y$ of the soccer field (in terms of a and b) that maximise its area. 2 marks

OR

(iii)(b) Use the second derivative test to find the length $2x$ and width $2y$ of the soccer field (in terms of a and b) that maximise its area. 2 marks

Formula / idea: A is maximum where $A'(x)=0$ and A' changes from $+$ to $-$ (or $A''<0$).

(i) 1 mark

1 A corner $P(x,y)$ lies on the ellipse
$y=\frac{b}{a}\sqrt{a^2-x^2}$ ✓ ½

2 Area $=2x\times 2y$
$A(x)=\frac{4b}{a}x\sqrt{a^2-x^2},\quad 0<x<a$ ✓ ½

Answer (i) $A(x)=\frac{4b}{a}x\sqrt{a^2-x^2}$

(ii) 1 mark

1 Differentiate w.r.t. x
$A'(x)=\frac{4b}{a}\left[\sqrt{a^2-x^2}-\frac{x^2}{\sqrt{a^2-x^2}}\right]=\frac{4b}{a}\cdot\frac{a^2-2x^2}{\sqrt{a^2-x^2}}$ ✓ ½

2 Put $A'(x) = 0$ $(x > 0)$ ✓ ½

$a^2 - 2x^2 = 0 \Rightarrow x = \dfrac{a}{\sqrt{2}}$

Answer (ii) **Critical point** $x = \dfrac{a}{\sqrt{2}}$

(iii)(a) 2 marks

1 Sign of $A'(x)$ depends on $a^2 - 2x^2$ ✓ 1

$0 < x < \frac{a}{\sqrt{2}} : A'(x) > 0, \qquad \frac{a}{\sqrt{2}} < x < a : A'(x) < 0$

2 A' changes from $+$ to $-$, so A is maximum at $x = \dfrac{a}{\sqrt{2}}$; then ✓ ½

$y = \dfrac{b}{a}\sqrt{a^2 - \dfrac{a^2}{2}} = \dfrac{b}{\sqrt{2}}$

3 Dimensions of the field ✓ ½

$2x = \sqrt{2}\,a, \qquad 2y = \sqrt{2}\,b$

Answer (iii)(a) **Length** $2x = \sqrt{2}\,a$**, width** $2y = \sqrt{2}\,b$

OR

(iii)(b) 2 marks

1 Differentiate $A'(x) = \dfrac{4b}{a}(a^2 - 2x^2)(a^2 - x^2)^{-1/2}$ again ✓ ½

$A''(x) = \dfrac{4b}{a}\left[\dfrac{-4x}{\sqrt{a^2 - x^2}} + \dfrac{x(a^2 - 2x^2)}{(a^2 - x^2)^{3/2}}\right]$

2 At $x = \dfrac{a}{\sqrt{2}}$: $a^2 - 2x^2 = 0$ and $\sqrt{a^2 - x^2} = \dfrac{a}{\sqrt{2}}$ ✓ ½

$A''\left(\frac{a}{\sqrt{2}}\right) = \dfrac{4b}{a} \cdot \dfrac{-4 \cdot \frac{a}{\sqrt{2}}}{\frac{a}{\sqrt{2}}} = -\dfrac{16b}{a} < 0$

3 So A is maximum at $x = \dfrac{a}{\sqrt{2}}$; then ✓ ½

$y = \dfrac{b}{a}\sqrt{a^2 - \dfrac{a^2}{2}} = \dfrac{b}{\sqrt{2}}$

4 Dimensions of the field ✓ ½

$2x = \sqrt{2}\,a, \qquad 2y = \sqrt{2}\,b$

Answer (iii)(b) **Length** $2x = \sqrt{2}\,a$**, width** $2y = \sqrt{2}\,b$

Watch out: Taking the area as xy; the field's length and breadth are $2x$ and $2y$, so $A = 4xy$.